Indications for ORIF

- Displaced **unfavourable fractures** (mandibular angle).
- Severely **atrophic edentulous mandibles**.
- Complex **facial fractures**.
- **Condylar fractures**.
- **Non-unions or malunions** after improper reduction.

Approaches for ORIF

- Wire osteosynthesis.
- **Intraoral approach**.
- **Submandibular approach**.
- **Retromandibular approach**.
- **Preauricular approach**.

Closed Reduction (CR)

- **Nondisplaced favourable fractures**.
- **Grossly comminuted fractures** (minimizes periosteum stripping).
- Fractures in **children** (protects tooth buds or developing dentition).
- **Coronoid fractures**: No treatment unless zygomatic arch impingement occurs.

Procedures for CR

- Erich arch bars.
- Bridle wires.
- Ivy loops.
- Biphasic pin fixation.

Causes of Hypothyroidism

- **Primary Hypothyroidism:**
  - Autoimmune (spontaneous atrophic).
  - **Iodine deficiency.**
  - **Post-131I therapy, thyroidectomy.**
  - **Hashimoto's thyroiditis.**
  - Genetic defects (dyshormonogenesis).
  - Drugs (lithium, amiodarone, etc.).
  - Radiation.
- **Secondary Hypothyroidism:**
  - Pituitary gland destruction (surgery, tumors).
  - Hypothalamic disorders.

Clinical Features

1. **General:** Weakness, cold intolerance, weight gain, puffy face, dry skin, hoarseness, goiter.
2. **GI**: Low appetite, constipation, ascites.
3. **Cardio**: Bradycardia, hypertension, cardiac failure.
4. **Neuromuscular**: Muscle stiffness, delayed reflexes, depression.
5. **Skin**: Dry flaky skin, alopecia, vitiligo, myxoedema.
6. **Reproductive**: Menorrhagia, infertility, impotence.
7. **Hematology**: Anemia, macrocytosis.
8. **Others**: Tiredness, slurred speech, cold intolerance.

Investigations

1. **TSH**: High in primary, low in secondary.
2. **T3/T4:** Low levels.
3. **Antibodies**: High TPO, thyroglobulin in Hashimoto's.
4. **Other labs:** High cholesterol, triglycerides, creatine kinase, hyponatremia.

Aetiology:

- **Environmental factors:**
  - Maternal epilepsy, **antiepileptic drugs** (phenytoin, diazepam), steroids, smoking, rubella.
  - **Alcohol, anticonvulsants, retinoic acid** linked to cleft formation.
- **Genetic factors:**
  - First-degree relatives have **25x higher risk.**
  - Common syndromes: **Pierre-Robin sequence, Stickler, Down syndrome, Apert, Treacher Collins.**

Pathophysiology:

- Lip and palate form via **mesenchymal migration and fusion.**
- Failure of **medial nasal and maxillary process fusion** → cleft lip.
- Failure of **palatal shelf fusion** → cleft palate.
- Causes: **Abnormal growth, cell death, or mesenchymal failure.**

Clinical Features:

- Visible **fissure or cleft** in lip/palate.
- **Misaligned teeth,** nasal distortion, speech difficulties.
- **Feeding problems** (milk leakage through nose).
- **Recurrent ear infections, poor growth.**

Management:
Cleft Lip:

- Repair done at **10 weeks** (Rule of 10s: **10 weeks, 10 pounds, 10 g hemoglobin**).
- Techniques:
  - **Millard rotation-advancement flap** (corrects lip & nasal deformity).
  - **Tennison-Randall repair:** Lengthens lip with triangular flap.

Cleft Palate:

- Repair done at **6-12 months.**
- Goals:
  - Separate **oral and nasal cavities** (airtight valve for normal speech).
  - Preserve **facial growth and dentition.**
- Techniques:
  - **Double-opposing Z-plasty** (realigns levator muscles).
  - **Straight-line repair** with muscle reorientation.
  - Closure of hard palate with **vomer flap.**

Aetiology

- **Known risk factors** include:
  - **Chronic hypoxic stimulation** (low oxygen levels over time).
  - **Genetic predisposition** (family history of similar tumours).

Clinical Features

- **Age of onset**: Most commonly around **45 years**.
- **Presentation**:
  - **Asymptomatic** mass in the **anterior triangle of the neck** (often the first sign).
  - Slow-growing, often **asymptomatic** for many years.
- **On examination**:
  - Mass is typically **vertically fixed** due to attachment to the **carotid bifurcation** (Fontaine sign).
  - **Bruit** (abnormal sound) may be heard when palpating the tumour.
  - **10%** of cases may present with **cranial nerve palsy** affecting:
    - **Hypoglossal nerve** (paralysis of the tongue).
    - **Glossopharyngeal nerve** (affecting swallowing and taste).
    - **Recurrent laryngeal nerve** (hoarseness).
    - **Spinal accessory nerve** (shoulder drop).
    - **Sympathetic chain** involvement (Horner syndrome: ptosis, miosis, anhidrosis).

Symptoms Associated with CBTs

- **Pain**
- **Hoarseness**
- **Dysphagia** (difficulty swallowing)
- **Horner syndrome** (drooping eyelid, constricted pupil)
- **Shoulder drop**

Signs and Symptoms of Shock

1. **CNS Symptoms:**
   - **Anxiety, restlessness, altered mental state** → ↓ cerebral perfusion and hypoxia.
2. **Cardiovascular Symptoms:**
   - **Hypotension** → ↓ circulatory volume.
   - **Tachycardia** with weak, **thready pulse** → ↓ blood flow.
3. **Skin Changes:**
   - **Cool, clammy skin** → vasoconstriction.
   - **Cold, mottled extremities** → poor perfusion.
   - **Pallor.**
4. **Respiratory Symptoms:**
   - **Rapid, shallow breathing (tachypnea)** → acidosis.
5. **Other Symptoms:**
   - **Hypothermia** → ↓ perfusion, sweat evaporation.
   - **Thirst and dry mouth** → fluid depletion.
   - **Fatigue** → inadequate oxygenation.
   - **Fainting** → ↓ blood supply.
   - **Oliguria/anuria** → ↓ renal perfusion, afferent arteriolar vasoconstriction.

E. Drugs in management of shock

1. **Sedatives:**
    - Relieve pain and anxiety.
    - **Morphine (IV)** is preferred; avoid IM due to delayed absorption in shock.
    - **Pethidine** is also effective.
2. **Chronotropic Agents:**
    - For low heart rate: Use **atropine** or **isoproterenol**.
3. **Inotropic Agents:**
    - Enhance cardiac muscle contraction (e.g., **dopamine, dobutamine**).
    - Benefits:
        - ↑ **Myocardial activity**.
        - ↑ **Renal blood flow** via vasodilation.
    - Use carefully in **small doses**.
4. **Vasodilators:**
    - For **septic, traumatic, cardiogenic shock**.
    - Examples: **Nitroprusside, nitroglycerin** (short-acting and reversible).
5. **Vasoconstrictors:**
    - Used in **neurogenic shock** to:
        - ↑ **BP** and **coronary circulation perfusion**.
        - ↑ **Myocardial activity**.
    - Examples: **Phenylephrine, metaraminol.**
    - **Avoid in hypovolemic and traumatic shock.**
6. **Beta-Blockers:**
    - For **stiff myocardium** and **rapid heart rate** in cardiogenic shock.
    - Example: **Propranolol**.
7. **Diuretics:**
    - For **cardiogenic shock** to ↓ vascular volume and filling pressure.
    - **Avoid in hypovolemic and septic shock**, as they worsen hypovolemia.

Clinical Features of tetanus

- Incubation: 4–14 days.
- **Symptoms:**
  - **Trismus (lockjaw):** 75% cases.
  - **Stiffness**: Starts in **jaw/neck** → spreads in 24–48 hrs.
  - **Spasms**: Triggered by noise/light → risk of **apnoea, fractures, asphyxia.**
  - **Risus sardonicus:** Sneering grin.
  - Severe: **Opisthotonus**, diaphragm spasms → **autonomic dysfunction.**

Diagnostic Test

- **Spatula test:** Reflex spasm → patient bites spatula.

Types

- Generalized: Most common; starts with trismus → spreads.
- **Neonatal**: Due to infected **umbilical stump.**
- **Local**: Spasms near injury site.
- **Cephalic**: Involves **cranial nerves.**

Aetiopathogenesis of solitary thyroid nodule

- **Thyroiditis**
- **Follicular adenoma** (most comm[illegible]se).
- **Thyroid cyst**
- **Colloid nodules**
- **Carcinoma**

Clinical Features

- Common in **females (20-40 years)**, 80% occur in women.
- **Euthyroid** in most cases.
- **Symptoms:**
    - Long-standing neck swelling.
    - **Dyspnoea, dysphagia.**
    - Obstructive signs (vein engorgement, stridor, tracheal deviation).
- Single nodule present.
- **Hard area:** Calcification.
- **Soft area:** Necrosis.
- **Sudden increase in size:** Possible haemorrhage.
- **Higher malignancy risk** than multi-nodular goitre.

→ Redness, swelling, heat & pus formation

Post-operative wound complication infection
[FESH² - DP- walk]

Management of shock: (↑CO & improve tissue perfusion)
1) Resuscitation [R-CE-CD]
2) Control of hemorrhage
3) Extracellular fluid replacement
4) Correct acid-base balance
5) Drugs

Treatment of septic shock: [IFVMS]
1) Inf. control
2) Fluid replacement
3) Vasopressor therapy
4) Vasoactive drugs
5) Mechanical ventilation
6) steroid therapy

clostridium tetani → anaerobic gram +ve bacillus

Management of tetanus :-
1) Diagnosis
Wound care
Antibiotics (metronidazole (IV) (10 days)
Passive immunization
Supportive care (Diazepam → spasm)

Universal precaution :
Precautions in OPD
[GDP- HR- W]
Gloves, Disposable ins, Pt management, Hand hygiene, Reusable ins, waste disposal.

→ Precautions in operating room : [E-DSP^2]
Enviornmental cleaning, Dental chair, Surgical tech, Personnel, Post-surgery inst. care

→ Post-exposure prophylaxis for health workers:
(1). Immediate action
(2) Prophylactic medication
ART, AZT (zidovudine)
(3). Follow up
(4). Vaccination

→ Prec. for air-borne diseases:
- wear N-95 masks

→ Prec. for COVID-19

~~# Indications for ORIF in mandibular frac:~~
~~(1) Displaced unfavourable frac. (mandibular angle)~~
~~(2)~~

Management of open fractures:-
(1). Wound care
(2) Antibiotics
(3). Tetanus prophylaxis
(4). Reduction & fixation
(5). Immobilization
(6). Rehabilitation